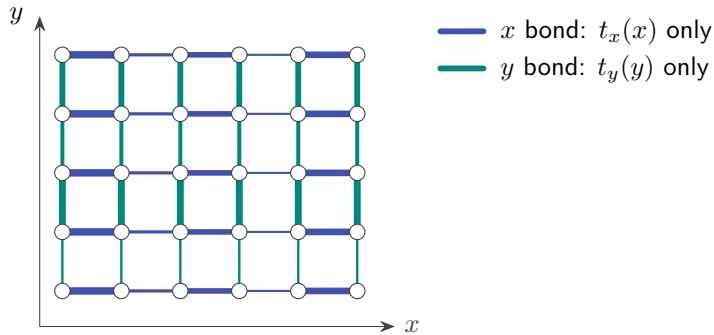


### (a) Separable 2D quasiperiodic hopping

$$H_0 = \sum_{\langle ij \rangle} t_{ij} (c_i^\dagger c_j + c_j^\dagger c_i)$$

$$t_x(x) = -1 - V_2 \cos\left[2\pi\tau\left(x + \frac{1}{2}\right)\right]$$

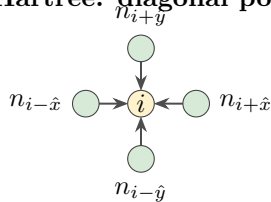
$$t_y(y) = -1 - V_2 \cos\left[2\pi\tau\left(y + \frac{1}{2}\right)\right]$$



Bond thickness depicts  $|t_{ij}|$ . The horizontal pattern repeats in every row; the vertical pattern repeats in every column.

### (b) Local Hartree/Fock fields

**Hartree: diagonal potential at  $i$**



$$(V_H)_{ii} = U \sum_{j \in \text{nn}(i)} n_j$$

**Fock: correction on a physical bond**



$$(V_F)_{ij} = -U \text{Re } \rho_{ij}$$